TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	80.5 ± 2.91
Correlation Stability	83.1 ± 1.22
Distribution Consistency	86.1 ± 1.92
Entropy Preservation	82.2 ± 1.83
Mean Z Score Drift	0.4 ± 0.10
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	98.2 ± 1.33
Feature Accuracy	73.6 ± 6.31
Novelty Score	0.9 ± 0.01

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Dataset Audit: indian_diabetes.csv

1. Dataset Portrait

Attribute	Value
Rows	768
Columns	9
Numeric features	9
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	google/gemma-4-e2b
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Pregnancies	3.845	2.984	-0.861	22.4%	Warn
Glucose	120.895	99.533	-21.362	17.7%	Warn
BloodPressure	69.105	78.972	+9.867	14.3%	Warn
SkinThickness	20.536	23.313	+2.776	13.5%	Warn
Insulin	79.799	53.485	-26.314	33.0%	Drift
BMI	31.993	28.372	-3.620	11.3%	Warn
DiabetesPedigreeFunction	0.472	0.455	-0.017	3.5%	OK
Age	33.241	26.497	-6.744	20.3%	Warn
Outcome	0.349	0.436	+0.088	25.1%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	80.5 ± 2.91
Correlation Stability (*2)	83.1 ± 1.22
Distribution Consistency (*3)	86.1 ± 1.92
Entropy Preservation (*4)	82.2 ± 1.83
Mean Z Score Drift (*5)	0.4 ± 0.10
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	98.2 ± 1.33
Feature Accuracy (*10)	73.6 ± 6.31

Novelty Score (*11)	0.9 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.
[2] (*13) Folds: Contextual constraint or model parameter.
[3] (*14) Context %: Contextual constraint or model parameter.
[4] (*15) Generation %: Contextual constraint or model parameter.
[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7493506493506492	0.5629870129870132	0.7344155844155844	-0.014935064935064923	
F1	0.6076604413342895	0.3368309070461554	0.5602498080593403	-0.0474106332749491	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.753896103896104	0.5707792207792208	0.7467532467532468	-0.007142857142857150	
F1	0.6155511296147973	0.3918510612034914	0.5920500608940001	-0.02350106872079611	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.746103896103896	0.5759740259740259	0.7324675324675324	-0.013636363636363646	*
F1	0.6159022356442937	0.4015968492383012 5	0.5896109471680978	-0.026291288476196033	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6675324675324675	0.5220779220779221	0.6454545454545454	-0.022077922077922085	
F1	0.4907509554720443	0.3764601340613491 5	0.5099154259503105	0.019164470478266117	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.772077922077922	0.5766233766233766	0.7272727272727273	-0.044805194805194806	*
F1	0.6262914445612621	0.4334933716334569	0.5173007071279148	-0.10899073743334724	*

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7564935064935064	0.5857142857142857	0.7331168831168832	-0.02337662337662339	*
F1	0.5733680734057942	0.3283194159683638	0.4883195533456835	-0.08504852006011068	*

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7292207792207792	0.5766233766233767	0.7194805194805194	-0.009740259740259749	
F1	0.5921520638892815	0.4194013188057003	0.5801400872191316	-0.012011976670149466	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7636363636363636	0.524025974025974	0.7389610389610389	-0.024675324675324684	
F1	0.6399852354635025	0.3915869413201914	0.5626362739546098	-0.07734896150889248	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.714935064935065	0.5597402597402598	0.6876623376623376	-0.02727272727272726	
F1	0.5748649566274735	0.4010593148213059	0.5318251008996254	-0.04303985572784801	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7545454545454545	0.5694805194805196	0.7428571428571428	-0.011688311688311692	
F1	0.5998259463389416	0.3843496549346338	0.5824504486933532	-0.017375497645588413	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunct	Age	Outcome
1	6.0	148.0	72.0	35.0	0.0	33.6	0.627	50.0	1.0
2	1.0	85.0	66.0	29.0	0.0	26.6	0.351	31.0	0.0
3	8.0	183.0	64.0	0.0	0.0	23.3	0.672	32.0	1.0
4	1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
5	0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	1.0
6	5.0	116.0	74.0	0.0	0.0	25.6	0.201	30.0	0.0
7	3.0	78.0	50.0	32.0	88.0	31.0	0.248	26.0	1.0
8	10.0	115.0	0.0	0.0	0.0	35.3	0.134	29.0	0.0
9	2.0	197.0	70.0	45.0	543.0	30.5	0.158	53.0	1.0
10	8.0	125.0	96.0	0.0	0.0	0.0	0.232	54.0	1.0
11	4.0	110.0	92.0	0.0	0.0	37.6	0.191	30.0	0.0
12	10.0	168.0	74.0	0.0	0.0	38.0	0.537	34.0	1.0
13	10.0	139.0	80.0	0.0	0.0	27.1	1.441	57.0	0.0
14	1.0	189.0	60.0	23.0	846.0	30.1	0.398	59.0	1.0
15	5.0	166.0	72.0	19.0	175.0	25.8	0.587	51.0	1.0
16	7.0	100.0	0.0	0.0	0.0	30.0	0.484	32.0	1.0
17	0.0	118.0	84.0	47.0	230.0	45.8	0.551	31.0	1.0
18	7.0	107.0	74.0	0.0	0.0	29.6	0.254	31.0	1.0
19	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0
20	1.0	115.0	70.0	30.0	96.0	34.6	0.529	32.0	1.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	5.0	124.0	76.0	25.0	8.0	17.885007733683178	0.06478941904042097	31.0	0.0	1.0
2	17.0	111.0	67.0	37.0	5.0	22.901308968197018	1.3977407356811034	45.0	0.0	1.0
3	2.0	107.0	90.0	38.0	27.0	54.267458210716688	0.24012906935325695	27.0	1.0	1.0
4	0.0	103.0	60.0	39.0	7.0	30.152604983786005	0.8867280845528567	31.0	1.0	1.0
5	4.0	59.0	99.0	63.0	15.0	17.079680406070408	1.1840108580473345	33.0	0.0	1.0
6	0.0	37.0	58.0	14.0	4.0	39.047929530664796	1.0255483500520477	20.0	0.0	1.0
7	8.0	110.0	76.0	0.0	0.0	27.8	0.237	58.0	0.0	nan
8	1.0	90.0	62.0	18.0	59.0	25.1	1.268	25.0	0.0	nan
9	2.0	56.0	56.0	28.0	45.0	24.2	0.332	22.0	0.0	nan
10	5.0	111.0	72.0	28.0	0.0	23.9	0.407	27.0	0.0	nan

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	2.0	131.0	96.0	-6.0	-9.0	22.760113704646967	0.4285411002355576	16.0	1.0	2.0
2	-1.0	175.0	85.0	4.0	370.0	35.352522657750551	0.3691832026936298	26.0	0.0	2.0
3	6.0	145.0	92.0	-1.0	-24.0	18.176618217639009	0.2237550626067786	21.0	1.0	2.0
4	1.0	92.0	50.0	20.0	-6.0	34.295958439672106	0.2716302666956060 3	20.0	0.0	2.0
5	1.0	91.0	68.0	-2.0	199.0	52.524351610353538	0.3990427605257327	17.0	1.0	2.0
6	2.0	193.0	81.0	-2.0	-12.0	28.705430972227564	0.5842775582997135	24.0	0.0	2.0
7	0.0	91.0	68.0	32.0	210.0	39.9	0.381	25.0	0.0	nan
8	3.0	173.0	82.0	48.0	465.0	38.4	2.137	25.0	1.0	nan
9	1.0	147.0	94.0	41.0	0.0	49.3	0.358	27.0	1.0	nan
10	2.0	107.0	74.0	30.0	100.0	33.6	0.404	23.0	0.0	nan

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	0.0	59.0	56.0	22.0	-10.0	18.188639557286933	0.07139298427263824	26.0	0.0	3.0
2	2.0	69.0	50.0	13.0	29.0	17.766032693645798	0.20751932109302015	31.0	0.0	3.0
3	8.0	74.0	48.0	16.0	61.0	25.565290814517958	0.24649875221980658	30.0	0.0	3.0
4	15.0	108.0	51.0	19.0	8.0	40.088208477999046	0.28107914688498071	43.0	1.0	3.0
5	4.0	98.0	64.0	-1.0	-6.0	21.371303121172666	0.09936379691856001	28.0	1.0	3.0
6	3.0	73.0	50.0	20.0	46.0	42.789799484265733	0.18186468598456984	44.0	1.0	3.0
7	5.0	158.0	84.0	41.0	210.0	39.4	0.395	29.0	1.0	nan
8	1.0	101.0	50.0	15.0	36.0	24.2	0.526	26.0	0.0	nan
9	0.0	109.0	88.0	30.0	0.0	32.5	0.855	38.0	1.0	nan
10	5.0	147.0	75.0	0.0	0.0	29.9	0.434	28.0	0.0	nan

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	0.0	147.0	75.0	41.0	104.0	30.679073689489858	0.98056710887209018	59.0	0.0	4.0
2	0.0	74.0	97.0	10.0	74.0	28.022001209956873	0.71794364746067031	73.0	1.0	4.0
3	1.0	151.0	83.0	21.0	0.0	13.421938587135309	0.98625685505568006	34.0	0.0	4.0
4	4.0	112.0	59.0	29.0	-18.0	40.235187879115667	0.07684770858777046	46.0	0.0	4.0
5	1.0	127.0	104.0	27.0	-197.0	29.939852162418511	0.93335835823475002	69.0	1.0	4.0
6	0.0	191.0	109.0	48.0	-5.0	19.153121431543827	0.84067744115976032	24.0	1.0	4.0
7	5.0	117.0	86.0	30.0	105.0	39.1	0.251	42.0	0.0	nan
8	3.0	74.0	68.0	28.0	45.0	29.7	0.293	23.0	0.0	nan
9	2.0	129.0	0.0	0.0	0.0	38.5	0.304	41.0	0.0	nan
10	1.0	126.0	56.0	29.0	152.0	28.7	0.801	21.0	0.0	nan

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	2.0	74.0	86.0	15.0	-4.0	32.61759737330753	0.16177026554392274	48.0	1.0	5.0
2	8.0	132.0	101.0	43.0	35.0	33.84876239664466	0.4456023890083738	28.0	1.0	5.0
3	4.0	91.0	59.0	-2.0	-13.0	47.51426937828226	0.7684506886316139	40.0	1.0	5.0
4	7.0	90.0	70.0	16.0	169.0	39.49693155544209	0.928129505847504	27.0	0.0	5.0
5	10.0	126.0	67.0	45.0	219.0	42.81194948554647	0.08881080740885483	54.0	0.0	5.0
6	4.0	85.0	98.0	3.0	441.0	49.150220518336965	0.14275126915149447	40.0	1.0	5.0
7	13.0	126.0	90.0	0.0	0.0	43.4	0.583	42.0	1.0	nan
8	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
9	1.0	181.0	78.0	42.0	293.0	40.0	1.258	22.0	1.0	nan
10	3.0	78.0	70.0	0.0	0.0	32.5	0.27	39.0	0.0	nan

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	77.0	59.0	33.0	13.0	35.402758259166006	0.3138025207098205	26.0	1.0	6.0
2	6.0	133.0	79.0	9.0	4.0	38.603735573746478	1.5805071349742315	42.0	0.0	6.0
3	6.0	241.0	72.0	9.0	-41.0	82.68592809361297	0.12233953089115604	52.0	1.0	6.0
4	13.0	104.0	83.0	24.0	-1.0	36.934852813547644	0.4534110942419482 5	26.0	0.0	6.0
5	6.0	158.0	75.0	39.0	-21.0	45.325844536838321	0.3709879090216721	32.0	1.0	6.0
6	2.0	115.0	54.0	11.0	17.0	24.721017597414768	0.2054372891855983 4	36.0	0.0	6.0
7	0.0	102.0	52.0	0.0	0.0	25.1	0.078	21.0	0.0	nan
8	4.0	76.0	62.0	0.0	0.0	34.0	0.391	25.0	0.0	nan
9	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0	nan
10	1.0	124.0	74.0	36.0	0.0	27.8	0.1	30.0	0.0	nan

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	6.0	133.0	77.0	53.0	163.0	19.711247246497866	0.4573428260692905	39.0	1.0	7.0
2	-1.0	205.0	75.0	13.0	89.0	21.02929765975837	0.20235587172541955	28.0	0.0	7.0
3	0.0	83.0	94.0	3.0	412.0	26.575979992919001	1.02666327440508 7	43.0	0.0	7.0
4	8.0	148.0	87.0	33.0	-23.0	40.687786707168016	0.23931263559257449	26.0	0.0	7.0
5	2.0	137.0	81.0	29.0	-27.0	40.461593542339 48	0.42052354747564 58	55.0	1.0	7.0
6	-2.0	144.0	51.0	2.0	189.0	36.691947107329554	0.84938479575887 2	41.0	0.0	7.0
7	2.0	85.0	65.0	0.0	0.0	39.6	0.93	27.0	0.0	nan
8	2.0	142.0	82.0	18.0	64.0	24.7	0.761	21.0	0.0	nan
9	9.0	102.0	76.0	37.0	0.0	32.9	0.665	46.0	1.0	nan
10	4.0	117.0	64.0	27.0	120.0	33.2	0.23	24.0	0.0	nan

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	139.0	54.0	-1.0	-11.0	45.056000049309475	0.24052042493550002	22.0	0.0	8.0
2	10.0	116.0	88.0	-2.0	-10.0	27.144711420212 17	0.09971367919975024	38.0	1.0	8.0
3	8.0	134.0	75.0	-3.0	596.0	37.313816077018 82	1.96266678159873 86	31.0	0.0	8.0
4	10.0	179.0	56.0	1.0	-3.0	9.86700417181633	0.66034207918404 39	18.0	0.0	8.0
5	7.0	125.0	72.0	-2.0	8.0	21.587882334783412	2.12380568753574 74	46.0	0.0	8.0
6	4.0	171.0	64.0	41.0	-20.0	26.764415466130124	0.41169845328847354	28.0	1.0	8.0
7	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
8	8.0	196.0	76.0	29.0	280.0	37.5	0.605	57.0	1.0	nan
9	1.0	118.0	58.0	36.0	94.0	33.3	0.261	23.0	0.0	nan
10	1.0	116.0	70.0	28.0	0.0	27.4	0.204	21.0	0.0	nan

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	3.0	180.0	93.0	37.0	6.0	33.5954350181659 5	0.37078313921208 83	23.0	0.0	9.0
2	3.0	86.0	77.0	51.0	65.0	15.0739382121311 87	0.45510821988790 434	34.0	0.0	9.0
3	3.0	108.0	76.0	44.0	322.0	23.7527158326937 17	0.70012865155175 41	33.0	0.0	9.0
4	6.0	137.0	101.0	6.0	258.0	37.7136623184257 05	0.32733296450696 414	26.0	1.0	9.0
5	13.0	184.0	93.0	33.0	121.0	27.2946273220405 15	0.88913889900393 37	19.0	1.0	9.0
6	4.0	142.0	109.0	4.0	3.0	28.1159682719345 02	-0.00598696569272 9274	34.0	0.0	9.0
7	8.0	65.0	72.0	23.0	0.0	32.0	0.6	42.0	0.0	nan
8	10.0	75.0	82.0	0.0	0.0	33.3	0.263	38.0	0.0	nan
9	2.0	146.0	70.0	38.0	360.0	28.0	0.337	29.0	1.0	nan
10	5.0	126.0	78.0	27.0	22.0	29.6	0.439	40.0	0.0	nan

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	-1.0	154.0	60.0	-2.0	-6.0	16.5329695092058 9	0.17351672006389 834	16.0	1.0	10.0
2	4.0	111.0	90.0	9.0	225.0	17.5965832263457 86	1.21097065753039 11	16.0	1.0	10.0
3	10.0	103.0	54.0	5.0	-5.0	19.8730721373831 26	0.47902023774283 914	22.0	0.0	10.0
4	-1.0	78.0	90.0	33.0	-8.0	32.2189008655449 4	0.53335509969779 95	28.0	1.0	10.0
5	7.0	205.0	90.0	28.0	140.0	14.9424530065488 86	0.35109794310137 6	19.0	1.0	10.0
6	9.0	93.0	61.0	27.0	-18.0	38.9612879112750 1	0.66261866733015 44	27.0	0.0	10.0
7	3.0	180.0	64.0	25.0	70.0	34.0	0.271	26.0	0.0	nan
8	3.0	99.0	80.0	11.0	64.0	19.3	0.284	30.0	0.0	nan
9	4.0	148.0	60.0	27.0	318.0	30.9	0.15	29.0	1.0	nan
10	7.0	136.0	90.0	0.0	0.0	29.9	0.21	50.0	0.0	nan